

TRENTOS-M RasPi4 port

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Scope

- RasPi4 port, basically a snapshot copy of LRZ GitLab content
- see [TRENTOS-M on RPi 4](#) for the porting plan

Tools

- https://www.raspberrypi.org/documentation/configuration/boot_folder.md
- binary blobs: <https://github.com/raspberrypi/firmware/tree/master/boot>
- misc docs
 - <https://www.golem.de/news/raspberry-pi-der-mit-dem-64-bit-kernel-tanz-1611-124475.html>
 - https://elinux.org/RPi_U-Boot

32-bit vs 64-bit

- source at sdk-sel4-camkes/...
 - kernel
 - include/arch/arm/arch/types.h
 - word_t = unsigned long
- ```
/* for libsel4 headers that the kernel shares */
typedef word_t sel4_Word;
```
- include/stdint.h
    - **uint64\_t = unsigned long long**
  - kernel/ libsel4
    - arch\_include/arm/sel4/arch/simple\_types.h
      - **sel4\_Uint32 = unsigned int**
    - libsel4/sel4\_arch\_include/<arch>/sel4/sel4\_arch/types.h
      - includes sel4/simple\_types.h -> libsel4/include/sel4/simple\_types.h
        - includes sel4/arch/simple\_types.h
        - include sel4/sel4\_arch/simple\_types.h -> libsel4/sel4\_arch\_include/<arch>/sel4/sel4\_arch/simple\_types.h
          - aarch32:
            - **sel4\_Uint64 = unsigned long long**
            - **missing: PRI\_sel4\_u64 = "ull"**
          - aarch64:
            - **sel4\_Uint64 = unsigned long**
            - **missing: PRI\_sel4\_u64 = "ul"**
    - aarch32:
      - **sel4\_Word = sel4\_Uint32**
      - **missing: PRI\_sel4\_word = "u"**
    - aarch64:
      - **sel4\_Word = sel4\_Uint64**
      - **missing: PRI\_sel4\_word = PRI\_sel4\_u64**
  - libs/sel4\_util\_libs
    - libutils/include/utils/time.h uses hard-coded "ull" postfixed which can conflict with the **uint64\_t** and **PRId64** definition from musllibc. The code

```
uint64_t timeout = 42000
Debug_LOG_INFO("v = %" PRIu64 " sec", timeout/MS_IN_S);
```

does not compile and the error is "integer type is 'ull', but format specifier is 'ul'". This can be explained as MS\_IN\_S is defined as "1000ull" and uint64\_t is "ul" (on 64-bit targets). According to C99 section 6.3.1.8 'Usual arithmetic conversions', the resulting expression has the type of the highest integer rank of the operands involved, so "ull" wins over "ul". The underlying bug is actually, that time.h should not hard-code "ull" there, but adapt to the uint64\_t definition of the runtime library.

- **libs/musllibc**

- **include/alltypes.h.in**

```
TYPEDEF unsigned _Addr size_t;
TYPEDEF unsigned _Addr uintptr_t;
TYPEDEF unsigned _Int64 uint64_t;
```

- **include/stdint.h**

```
#define __NEED_uint64_t
#include <bits/alltypes.h>
```

- **include/sys/types.h**

```
#if defined(_GNU_SOURCE) || defined(_BSD_SOURCE)
#define __NEED_int64_t
#endif
#include <bits/alltypes.h>
```

- **include/inttypes.h**

```
#include <stdint.h>
#include <bits/alltypes.h>

#if UINTPTR_MAX == UINT64_MAX
#define __PRI64 "l"
#define __PRIPTR "l"
#else
#define __PRI64 "ll"
#define __PRIPTR ""
#endif

#define PRId64 __PRI64 "d"
#define PRIi64 __PRI64 "i"
#define PRIu64 __PRI64 "u"
#define PRIx64 __PRI64 "x"
```

- **build**

- **<build-32>/musllibc/build-temp/obj/include/bits/alltypes.h**

- ??? intermediate artifact 

- **<build-32>/musllibc/build-temp/stage/include/bits/alltypes.h**

- this is included by the build due to " -Imusllibc/build-temp/stage/include "
    - 32-bit

```
#define _Addr int
#define _Int64 long long

#if defined(__NEED_uint64_t) && !defined(__DEFINED_uint64_t)
typedef unsigned _Int64 uint64_t;
#define __DEFINED_uint64_t
#endif
```

- **PRIu64** is "%llu" (since **UINTPTR\_MAX** is **UINT32\_MAX** from **sdk-sel4-camkes/libs/musllibc/arch/arm/bits/stdint.h** )
      - **uint64\_t** is **long long unsigned int**

- 64-bit

```
#define _Addr long
#define _Int64 long

#if defined(__NEED_uint64_t) && !defined(__DEFINED_uint64_t)
typedef unsigned _Int64 uint64_t;
#define __DEFINED_uint64_t
#endif
```

- **PRIu64** is "%lu" (since **UINTPTR\_MAX** is **UINT64\_MAX** from **sdk-sel4-camkes/libs/musllibc/arch/aarch64\_sel4/bits/stdint.h**)
- **uint64\_t** is **long unsigned int**

## EEPROM

- SPI EEPROM, 4 MiBit (=512 KiByte)
  - content are executed by BCM2711 ROM during boot, unless a file `recovery.bin` exists in the SD card which take precedence over the ROM, allows unbricking
  - <https://www.raspberrypi.org/documentation/hardware/raspberrypi/booteeeprom.md>

## U-Boot build process

The final U-Boot image used in the project is <https://github.com/apritzel/u-boot/tree/rpi4-eth-v3> as this version offers support for the new Gigabit Ethernet (GENET) controller on the RPi4. The source code can be obtained with:

```
git clone https://github.com/apritzel/u-boot.git
cd <path/to/u-boot>
git checkout rpi4-eth-v3
```

According to <https://docs.sel4.systems/Hardware/Rpi3.html>, U-Boot for the RPi3 must disable caches due to cache configuration issues in the sel4 ELFLoader. This can be done by reverting commit <https://github.com/u-boot/u-boot/commit/995eab8b5b580b67394312b1621c60a71042cd18>. For RPi4 this seems not necessary. For the RPi4 in 32-bit mode, it is necessary though:

```
git revert 995eab8b5b580b67394312b1621c60a71042cd18
```

Compile U-Boot binary for RPi4 in 32-bit mode (aarch64-linux-gnu-) with:

```
make CROSS_COMPILE=arm-linux-gnueabi- rpi_4_32b_defconfig
make CROSS_COMPILE=arm-linux-gnueabi- -j8
```

Compile U-Boot binary for RPi4 in 64-bit mode (aarch64-linux-gnu-) with:

```
make CROSS_COMPILE=aarch64-linux-gnu- rpi_4_defconfig
make CROSS_COMPILE=aarch64-linux-gnu- -j8
```

## config.txt

Contains many configuration parameters for setting up the Pi. Current configuration for 32-bit platform:

```
uart_2ndstage=1
enable_uart=1
kernel=u-boot.bin
```

Current configuration for 64-bit platform:

```
arm_64bit=1
uart_2ndstage=1
enable_uart=1
kernel=u-boot.bin
```

## boot.scr

boot.scr is a user-defined image file that is read before loading the seL4 kernel image, e.g. os\_image.bin. Create boot.cmd file:

```
fatload mmc 0 0x10000000 os_image.bin
bootelf 0x10000000
```

Create boot.scr for 32-bit platform from boot.cmd file:

```
mkimage -A arm -T script -C none -n "Boot script" -d "boot.cmd" boot.scr
```

Create boot.scr for 64-bit platform from boot.cmd file:

```
mkimage -A arm64 -T script -C none -n "Boot script" -d "boot.cmd" boot.scr
```

## seL4test

```
Initialising SDRAM 'Micron' 32Gb x2 total-size: 64 Gbit 3200
recover4.elf not found
recovery.elf not found
Read start4.elf bytes 2212384 hnd 0x0000001f hash bf07104c7821ae07
Read fixup4.dat bytes 5421 hnd 0x0000001c hash 4e7be95384e3e746
0x00d03114 0x00000000 0x0000001f
MEM GPU: 76 ARM: 948 TOTAL: 1024
FIXUP src: 128 256 dst: 948 1024
Starting start4.elf @ 0xfec00200

U-Boot 2020.01-gea7ab14e67 (Nov 24 2020 - 21:22:28 +0100)

DRAM: 3.9 GiB
RPI 4 Model B (0xd03114)
MMC: mmc0@7e300000: 1, emmc2@7e340000: 0
Loading Environment from FAT... *** Warning - bad CRC, using default environment

In: serial
Out: serial
Err: serial
Net: eth0: ethernet@7d580000
Hit any key to stop autoboot: 2 1 0
switch to partitions #0, OK
mmc0 is current device
Scanning mmc 0:1...
Found U-Boot script /boot.scr
129 bytes read in 8 ms (15.6 KiB/s)
Executing script at 02400000
5379752 bytes read in 359 ms (14.3 MiB/s)
Starting application at 0x00a56000 ...

ELF-loader started on CPU: ARM Ltd. Cortex-A58 r0p3
paddr=[a56000..f590f7]
No DTB passed in from boot loader.
Looking for DTB in CPIO archive...found at b7fb10.
Loaded DTB from b7fb10.
paddr=[23f000..245fff]
ELF-loading image 'kernel'
paddr=[0..23efff]
```

```

vaddr=[ffffff8000000000..ffffff800023efff]
virt_entry=ffffff8000000000
ELF-loading image 'sel4test-driver'
paddr=[246000..65cfff]
vaddr=[400000..816fff]
virt_entry=40f0b8
Enabling MMU and paging
Jumping to kernel-image entry point...

```

```

Bootstrapping kernel
Booting all finished, dropped to user space
Node 0 of 1
IOPT levels: 0
IPC buffer: 0x817000
Empty slots: [1169 --> 8192)
sharedFrames: [0 --> 0)
userImageFrames: [27 --> 1074)
userImagePaging: [14 --> 19)
untypedes: [1074 --> 1169)
Initial thread domain: 0
Initial thread cnode size: 13
List of untypedes

```

```

Paddr | Size | Device
0x80000000 | 27 | 1
0x10000000 | 28 | 1
0x20000000 | 29 | 1
0x40000000 | 30 | 1
0x80000000 | 30 | 1
0xc0000000 | 29 | 1
0xe0000000 | 28 | 1
0xf0000000 | 27 | 1
0xf8000000 | 26 | 1
0xfc000000 | 25 | 1
0xfe000000 | 24 | 1
0xff000000 | 23 | 1
0xff801000 | 12 | 1
0xff802000 | 13 | 1
0xff804000 | 14 | 1
0xff808000 | 15 | 1
0xff810000 | 16 | 1
0xff820000 | 17 | 1
0xff840000 | 12 | 1
0xff843000 | 12 | 1
0xff844000 | 14 | 1
0xff848000 | 15 | 1
0xff850000 | 16 | 1
0xff860000 | 17 | 1
0xff880000 | 19 | 1
0xff900000 | 20 | 1
0xffa00000 | 21 | 1
0xffc00000 | 22 | 1
0x100000000 | 32 | 1
0x200000000 | 33 | 1
0x400000000 | 34 | 1
0x800000000 | 35 | 1
0x1000000000 | 36 | 1
0x2000000000 | 37 | 1
0x4000000000 | 38 | 1
0x8000000000 | 43 | 1
0x88000000000 | 42 | 1
0xc8000000000 | 41 | 1
0xe8000000000 | 40 | 1
0xf8000000000 | 38 | 1
0xfc000000000 | 37 | 1
0xfe000000000 | 36 | 1
0xff000000000 | 35 | 1
0xff800000000 | 34 | 1
0xffc00000000 | 33 | 1
0xffe00000000 | 32 | 1
0xffff0000000 | 31 | 1

```

```

0xffff80000000 | 30 | 1
0xffffc0000000 | 29 | 1
0xffffe0000000 | 28 | 1
0xfffff0000000 | 27 | 1
0xfffff8000000 | 26 | 1
0xfffffc000000 | 25 | 1
0xfffffe000000 | 24 | 1
0xffffff000000 | 23 | 1
0xffffff800000 | 22 | 1
0xffffffc00000 | 21 | 1
0xffffffe00000 | 20 | 1
0xfffffff00000 | 19 | 1
0xfffffff80000 | 18 | 1
0xfffffffc0000 | 17 | 1
0xfffffff00000 | 16 | 1
0xfffffff00000 | 15 | 1
0xfffffff80000 | 14 | 1
0xfffffff00000 | 13 | 1
0xfffffff00000 | 12 | 1
0xfffffff00000 | 11 | 1
0xfffffff80000 | 10 | 1
0xfffffff00000 | 9 | 1
0xfffffff00000 | 8 | 1
0xfffffff00000 | 7 | 1
0xfffffff80000 | 6 | 1
0xfffffff00000 | 5 | 1
0xfffffff00000 | 4 | 1
0 | 16 | 0
0x65d000 | 12 | 0
0x65e000 | 13 | 0
0x660000 | 17 | 0
0x680000 | 19 | 0
0x700000 | 20 | 0
0x800000 | 23 | 0
0x1000000 | 24 | 0
0x2000000 | 25 | 0
0x4000000 | 25 | 0
0x6000000 | 24 | 0
0x7000000 | 23 | 0
0x7800000 | 22 | 0
0x7c00000 | 21 | 0
0x7e00000 | 20 | 0
0x7f00000 | 19 | 0
0x7fd1800 | 11 | 0
0x7fd2000 | 13 | 0
0x7fd4000 | 14 | 0
0x7fd8000 | 15 | 0
0x7fe0000 | 17 | 0
Untyped summary
1 untyped of size 4
1 untyped of size 5
1 untyped of size 6
1 untyped of size 7
1 untyped of size 8
1 untyped of size 9
1 untyped of size 10
2 untyped of size 11
5 untyped of size 12
4 untyped of size 13
4 untyped of size 14
4 untyped of size 15
4 untyped of size 16
5 untyped of size 17
1 untyped of size 18
4 untyped of size 19
4 untyped of size 20
3 untyped of size 21
3 untyped of size 22
4 untyped of size 23
4 untyped of size 24
4 untyped of size 25

```

```

2 untypededs of size 26
3 untypededs of size 27
3 untypededs of size 28
3 untypededs of size 29
3 untypededs of size 30
1 untypededs of size 31
2 untypededs of size 32
2 untypededs of size 33
2 untypededs of size 34
2 untypededs of size 35
2 untypededs of size 36
2 untypededs of size 37
2 untypededs of size 38
1 untypededs of size 40
1 untypededs of size 41
1 untypededs of size 42
1 untypededs of size 43
Switching to a safer, bigger stack...
sel4 Test
=====

vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 2147483648, error 1
vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 1073741824, error 1
vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 536870912, error 1
vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 268435456, error 1
vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 134217728, error 1
vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 67108864, error 1
vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 33554432, error 1
vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 16777216, error 1
vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 8388608, error 1
vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 4194304, error 1
vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 2097152, error 1
vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 1048576, error 1
vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 524288, error 1
vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 262144, error 1
vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 131072, error 1
vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 65536, error 1
vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 32768, error 1
vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 16384, error 1
vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 8192, error 1
vspace_reserve_range_at@vspace.h:738 vspace is NULL
create_reservations@elf.c:243 Failed to make reservation: 0x400000, 901120
elf_reserve_regions_in_vspace@elf.c:436 Failed to create reservations
sel4utils_elf_reserve@elf.c:463 Failed to reserve regions
Starting test suite sel4test
Starting test 0: Test that there are tests
Starting test 1: SYSCALL0000
Starting test 2: SYSCALL0001
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404c64]: CNodeCap: Illegal Operation attempted.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404c64]: CNodeCap: Illegal Operation attempted.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404c64]: CNodeCap: Illegal Operation attempted.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404c64]: CNodeCap: Illegal Operation attempted.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404c64]: CNodeCap: Illegal Operation attempted.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404c64]: CNodeCap: Illegal Operation attempted.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404c64]: CNodeCap: Illegal Operation attempted.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404c64]: CNodeCap: Illegal Operation attempted.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404c64]: CNodeCap: Illegal Operation attempted.>>[0m
Starting test 3: SYSCALL0002
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404c64]: CNodeCap: Illegal Operation attempted.>>[0m

```







```

[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404c64]: CNodeCap: Illegal Operation attempted.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404c64]: CNodeCap: Illegal Operation attempted.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404c64]: CNodeCap: Illegal Operation attempted.>>[0m
Starting test 14: SYSCALL0016
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [handleReply/498 T0xffffffff8007fd1400 "sel4test-driver" @404ca8]:
Attempted reply operation when no reply cap present.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [handleReply/498 T0xffffffff8007fd1400 "sel4test-driver" @404ca8]:
Attempted reply operation when no reply cap present.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [handleReply/498 T0xffffffff8007fd1400 "sel4test-driver" @404ca8]:
Attempted reply operation when no reply cap present.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [handleReply/498 T0xffffffff8007fd1400 "sel4test-driver" @404ca8]:
Attempted reply operation when no reply cap present.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [handleReply/498 T0xffffffff8007fd1400 "sel4test-driver" @404ca8]:
Attempted reply operation when no reply cap present.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [handleReply/498 T0xffffffff8007fd1400 "sel4test-driver" @404ca8]:
Attempted reply operation when no reply cap present.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [handleReply/498 T0xffffffff8007fd1400 "sel4test-driver" @404ca8]:
Attempted reply operation when no reply cap present.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [handleReply/498 T0xffffffff8007fd1400 "sel4test-driver" @404ca8]:
Attempted reply operation when no reply cap present.>>[0m
Starting test 15: SYSCALL0017
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404db4]: CNodeCap: Illegal Operation attempted.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404db4]: CNodeCap: Illegal Operation attempted.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404db4]: CNodeCap: Illegal Operation attempted.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404db4]: CNodeCap: Illegal Operation attempted.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404db4]: CNodeCap: Illegal Operation attempted.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404db4]: CNodeCap: Illegal Operation attempted.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404db4]: CNodeCap: Illegal Operation attempted.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404db4]: CNodeCap: Illegal Operation attempted.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/54 T0xffffffff8007fd1400 "sel4test-driver"
@404db4]: CNodeCap: Illegal Operation attempted.>>[0m
Starting test 16: TIMER0001
Starting test 17: TIMER0002
Starting test 18: BIND0001
Running test BIND0001 (Test that a bound tcb waiting on a sync endpoint receives normal sync ipc and
notification notifications.)
Test BIND0001 passed
Starting test 19: BIND0002
Running test BIND0002 (Test that a bound tcb waiting on its bound notification recieves notifications)
Test BIND0002 passed
Starting test 20: BIND0003
Running test BIND0003 (Test IPC ordering 1) bound tcb waits on bound notification 2, true) another tcb sends a
message)
Test BIND0003 passed
Starting test 21: BIND0004
Running test BIND0004 (Test IPC ordering 2) bound tcb waits on bound notification 1, true) another tcb sends a
message)
Test BIND0004 passed
Starting test 22: CACHEFLUSH0001
Running test CACHEFLUSH0001 (Test a cache maintenance on pages)
Test CACHEFLUSH0001 passed
Starting test 23: CACHEFLUSH0002
Running test CACHEFLUSH0002 (Test a cache maintenance on page directories)
Test CACHEFLUSH0002 passed

```

```

Starting test 24: CACHEFLUSH0003
Running test CACHEFLUSH0003 (Test that cache maintenance can be done on large pages)
Test CACHEFLUSH0003 passed
Starting test 25: CANCEL_BADGED_SENDS_0001
Running test CANCEL_BADGED_SENDS_0001 (Basic endpoint cancelBadgedSends testing.)
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff800066dc00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #207.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000682400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #268.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000682c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #331.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80006a8400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #403.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80006a8c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #464.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80006d0400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #526.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80006d0c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #583.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80006f6400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #655.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80006f6c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #709.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8007f1c400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #782.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8007f1cc00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #844.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8007f41400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #908.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8007f41c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #975.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8007f66400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1044.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8007f66c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1106.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff800070c400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1171.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff800070cc00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1230.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000732400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1293.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000732c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1355.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000757400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1429.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000757c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1490.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff800077c400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1563.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff800077cc00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1621.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80007a2400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1686.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80007a2c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1747.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80007c8400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1815.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80007c8c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1878.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80007ed400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1947.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80007edc00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1886.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8007e13400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #2070.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8007e13c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #2136.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8007e38400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #2081.>>[0m
Test CANCEL_BADGED_SENDS_0001 passed
Starting test 26: CANCEL_BADGED_SENDS_0002

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Attempted to invoke a null cap #270.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000682c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #332.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80006a8400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #399.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80006a8c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #461.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80006d1400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #506.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80006d1c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #572.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80006f6400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #639.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80006f6c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #693.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8007f1c400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #757.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8007f1cc00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #820.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8007f42400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #877.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8007f42c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #941.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8007f67400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1005.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8007f67c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1070.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff800070d400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1124.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff800070dc00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1192.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000733400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1371.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000733c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1433.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000758400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1373.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000758c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1560.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff800077e400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1625.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff800077ec00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1679.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80007a4400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1743.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80007a4c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1808.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80007c9400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1864.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80007c9c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1932.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80007ef400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #1992.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff80007efc00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #2047.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8007e15400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #2113.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8007e15c00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #2173.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8007e3a400 "child of: '27'" @41190c]:
Attempted to invoke a null cap #2237.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8007e3ac00 "child of: '27'" @41190c]:
Attempted to invoke a null cap #2298.>>[0m
Test CANCEL_BADGED_SENDS_0002 passed
Starting test 27: CNODEOP0001
Running test CNODEOP0001 (Basic seL4_CNode_Copy() testing)
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/95 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Destination not empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy

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[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/95 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Destination not empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
Test CNODEOP0007 passed
Starting test 34: CNODEOP0008
Running test CNODEOP0008 (Basic seL4_CNode_Rotate() testing)
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/95 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Destination not empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/95 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Destination not empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/95 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Destination not empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/95 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Destination not empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/95 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Destination not empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/266 T0xffffffff80041f0400 "27" @4007b8]: CNode
Rotate: Pivot slot the same as source or dest slot.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/95 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Destination not empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
Test CNODEOP0008 passed
Starting test 35: CNODEOP0009
Running test CNODEOP0009 (Basic seL4_CNode_SaveCaller() testing)
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [invokeCNodeSaveCaller/374 T0xffffffff80041f0400 "27" @4007b8]: CNode
SaveCaller: Reply cap not present.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/209 T0xffffffff80041f0400 "27" @4007b8]: CNode
SaveCaller: Destination slot not empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/95 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Destination not empty.>>[0m
Test CNODEOP0009 passed
Starting test 36: CSPACE0001
Running test CSPACE0001 (Test full cspace resolution)
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #0.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #1.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #2.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #3.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #4.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #5.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #6.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #7.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #8.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #9.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #10.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #11.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #12.>>[0m

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[illegible]



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Attempted to invoke a null cap #6184943488255481173.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #6076857097198589269.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #6293029879312373077.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #5860684315084805461.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #6725375443539940693.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #4995993186629670229.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #8454757700450211157.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #1537228672809129301.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/633 T0xffffffff8000673c00 "child of: '27'" @40f378]:
Attempted to invoke a null cap #15372286728091293013.>>[0m
Test CSPACE0001 passed
Starting test 37: DOMAINS0001
Running test DOMAINS0001 (Change domain successfully())
Test DOMAINS0001 passed
Starting test 38: DOMAINS0002
Running test DOMAINS0002 (Try non-existent domain())
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeDomainInvocation/1565 T0xffffffff80041f0400 "27" @410c70]: Domain
Configure: invalid domain (11 >= 1).>>[0m
Test DOMAINS0002 passed
Starting test 39: DOMAINS0003
Running test DOMAINS0003 (Invoke non-domain cap())
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeTCBInvocation/870 T0xffffffff80041f0400 "27" @410c70]: TCB: Illegal
operation.>>[0m
Test DOMAINS0003 passed
Starting test 40: DOMAINS0004
Running test DOMAINS0004 (Run threads in domains())
Test DOMAINS0004 passed
Starting test 41: FPU0000
Running test FPU0000 (Ensure that simple FPU operations work)
Test FPU0000 passed
Starting test 42: FPU0001
Running test FPU0001 (Ensure multiple threads can use FPU simultaneously)
Test FPU0001 passed
Starting test 43: FRAMEDIPC0001
Running test FRAMEDIPC0001 (Test that we cannot create a thread with an IPC buffer that is a frame)
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @41a6c0]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [checkValidIPCBuffer/621 T0xffffffff80041f0400 "27" @41a6c0]: IPC Buffer
is an invalid cap.>>[0m
Test FRAMEDIPC0001 passed
Starting test 44: FRAMEDIPC0002
Running test FRAMEDIPC0002 (Test that we cannot switch a threads IPC buffer to a device frame)
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @41a6c0]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [checkValidIPCBuffer/621 T0xffffffff80041f0400 "27" @41a6c0]: IPC Buffer
is an invalid cap.>>[0m
Test FRAMEDIPC0002 passed
Starting test 45: FRAMEDIPC0003
Running test FRAMEDIPC0003 (Test that deleting a frame cap unmaps the frame)
Test FRAMEDIPC0003 passed
Starting test 46: FRAMEEXPORTS0001
Running test FRAMEEXPORTS0001 (Test that we can access all exported frames)
vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 2097152, error 1
vka_alloc_object_at_maybe_dev@object.h:57 Failed to allocate object of size 4096, error 1
Test FRAMEEXPORTS0001 passed
Starting test 47: FRAMEXN0001
Running test FRAMEXN0001 (Test that we can map a small frame XN)
Test FRAMEXN0001 passed
Starting test 48: FRAMEXN0002
Running test FRAMEXN0002 (Test that we can map a large frame XN)
Test FRAMEXN0002 passed
Starting test 49: IPC0001
Running test IPC0001 (Test SMP seL4_Send + seL4_Recv)
Test IPC0001 passed

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Starting test 50: IPC0002
Running test IPC0002 (Test SMP seL4_Call + seL4_ReplyRecv)
Test IPC0002 passed
Starting test 51: IPC0003
Running test IPC0003 (Test SMP seL4_Send + seL4_Reply + seL4_Recv)
Test IPC0003 passed
Starting test 52: IPC0004
Running test IPC0004 (Test seL4_NBSend + seL4_Recv)
Test IPC0004 passed
Starting test 53: IPC0010
Running test IPC0010 (Test suspending an IPC mid-Call())
Test IPC0010 passed
Starting test 54: IPC1001
Running test IPC1001 (Test SMP inter-AS seL4_Send + seL4_Recv)
Test IPC1001 passed
Starting test 55: IPC1002
Running test IPC1002 (Test SMP inter-AS seL4_Call + seL4_ReplyRecv)
Test IPC1002 passed
Starting test 56: IPC1003
Running test IPC1003 (Test SMP inter-AS seL4_Send + seL4_Reply + seL4_Recv)
Test IPC1003 passed
Starting test 57: IPC1004
Running test IPC1004 (Test inter-AS seL4_NBSend + seL4_Recv)
Test IPC1004 passed
Starting test 58: IPCRIGHTS0001
Running test IPCRIGHTS0001 (seL4_Send needs write)
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/646 T0xffffffff80041f0400 "27" @426fe8]: Attempted to
invoke a read-only endpoint cap #208.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeInvocation/646 T0xffffffff80041f0400 "27" @426fe8]: Attempted to
invoke a read-only endpoint cap #208.>>[0m
Test IPCRIGHTS0001 passed
Starting test 59: IPCRIGHTS0002
Running test IPCRIGHTS0002 (seL4_Recv needs read)
Test IPCRIGHTS0002 passed
Starting test 60: IPCRIGHTS0003
Running test IPCRIGHTS0003 (seL4_Send with caps needs grant)
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff800066dc00 "child of: '27'"
@4007b8]: CNode Copy/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff800066dc00 "child of: '27'"
@4007b8]: CNode Copy/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
Test IPCRIGHTS0003 passed
Starting test 61: IPCRIGHTS0004
Running test IPCRIGHTS0004 (seL4_Call needs grant or grant-reply)
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [invokeCNodeSaveCaller/374 T0xffffffff80041f0400 "27" @4007b8]: CNode
SaveCaller: Reply cap not present.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/95 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Destination not empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [invokeCNodeSaveCaller/374 T0xffffffff80041f0400 "27" @4007b8]: CNode
SaveCaller: Reply cap not present.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/95 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Destination not empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [invokeCNodeSaveCaller/374 T0xffffffff80041f0400 "27" @4007b8]: CNode
SaveCaller: Reply cap not present.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m

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[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/95 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Destination not empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [invokeCNodeSaveCaller/374 T0xffffffff80041f0400 "27" @4007b8]: CNode
SaveCaller: Reply cap not present.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/95 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Destination not empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [invokeCNodeSaveCaller/374 T0xffffffff80041f0400 "27" @4007b8]: CNode
SaveCaller: Reply cap not present.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/95 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Destination not empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
Test IPCRIGHTS0004 passed
Starting test 62: IPCRIGHTS0005
Running test IPCRIGHTS0005 (seL4_Reply grant depends of the grant of previous seL4_Recv)
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff800066dc00 "child of: '27'"
@4007b8]: CNode Copy/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff800066dc00 "child of: '27'"
@4007b8]: CNode Copy/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/95 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Destination not empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff800066dc00 "child of: '27'"
@4007b8]: CNode Copy/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/95 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Destination not empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @4007b8]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff800066dc00 "child of: '27'"
@4007b8]: CNode Copy/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff800066dc00 "child of: '27'"
@4007b8]: CNode Copy/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff800066dc00 "child of: '27'"
@4007b8]: CNode Copy/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
Test IPCRIGHTS0005 passed
Starting test 63: NBWAIT0001
Running test NBWAIT0001 (Test seL4_NBRecv)
Test NBWAIT0001 passed
Starting test 64: PAGEFAULT0001
Running test PAGEFAULT0001 (Test read page fault)
Test PAGEFAULT0001 passed
Starting test 65: PAGEFAULT0002
Running test PAGEFAULT0002 (Test write page fault)
Test PAGEFAULT0002 passed
Starting test 66: PAGEFAULT0003
Running test PAGEFAULT0003 (Test execute page fault)
Test PAGEFAULT0003 passed
Starting test 67: PAGEFAULT0004
Running test PAGEFAULT0004 (Test unknown system call)
Test PAGEFAULT0004 passed
Starting test 68: PAGEFAULT0005
Running test PAGEFAULT0005 (Test undefined instruction)
Test PAGEFAULT0005 passed
Starting test 69: PAGEFAULT1001
Running test PAGEFAULT1001 (Test read page fault (inter-AS))
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @46776c]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
sel4utils_copy_path_to_process@process.c:131 Failed to copy cap

[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @46776c]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
sel4utils_copy_path_to_process@process.c:131 Failed to copy cap

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[illegible]



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Test PAGEFAULT1003 passed
Starting test 72: PAGEFAULT1004
Running test PAGEFAULT1004 (Test unknown system call (inter-AS))
[0m[30;1m<<[0m[32mseL4(CPU 0)][0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @46776c]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
sel4utils_copy_path_to_process@process.c:131 Failed to copy cap

[0m[30;1m<<[0m[32mseL4(CPU 0)][0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @46776c]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
sel4utils_copy_path_to_process@process.c:131 Failed to copy cap

[0m[30;1m<<[0m[32mseL4(CPU 0)][0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @46776c]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
sel4utils_copy_path_to_process@process.c:131 Failed to copy cap

[0m[30;1m<<[0m[32mseL4(CPU 0)][0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @46776c]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
sel4utils_copy_path_to_process@process.c:131 Failed to copy cap

[0m[30;1m<<[0m[32mseL4(CPU 0)][0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @46776c]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
sel4utils_copy_path_to_process@process.c:131 Failed to copy cap

[0m[30;1m<<[0m[32mseL4(CPU 0)][0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @46776c]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
sel4utils_copy_path_to_process@process.c:131 Failed to copy cap

[0m[30;1m<<[0m[32mseL4(CPU 0)][0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @46776c]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
sel4utils_copy_path_to_process@process.c:131 Failed to copy cap

[0m[30;1m<<[0m[32mseL4(CPU 0)][0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @46776c]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
sel4utils_copy_path_to_process@process.c:131 Failed to copy cap

[0m[30;1m<<[0m[32mseL4(CPU 0)][0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @46776c]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
sel4utils_copy_path_to_process@process.c:131 Failed to copy cap

[0m[30;1m<<[0m[32mseL4(CPU 0)][0m[30;1m [decodeCNodeInvocation/107 T0xffffffff80041f0400 "27" @46776c]: CNode Copy
/Mint/Move/Mutate: Source slot invalid or empty.>>[0m
sel4utils_copy_path_to_process@process.c:131 Failed to copy cap

Test PAGEFAULT1004 passed
Starting test 73: PREEMPT_REVOKE
Running test PREEMPT_REVOKE (Test preemption path in revoke)
Test PREEMPT_REVOKE passed
Starting test 74: PT0001
Running test PT0001 (Fun with page tables on ARM)
[0m[30;1m<<[0m[32mseL4(CPU 0)][0m[30;1m [decodeARMFrameInvocation/2129 T0xffffffff80041f0400 "27" @42f55c]:
ARMPageMap: Attempting to map frame into multiple addresses>>[0m
clear_entries_mid@vspace_internal.h:169 Cannot clear reserved entries mid level
Test PT0001 passed
Starting test 75: REGRESSIONS0001
Running test REGRESSIONS0001 (Ensure WriteRegisters functions correctly)
Test REGRESSIONS0001 passed
Starting test 76: REGRESSIONS0002
Running test REGRESSIONS0002 (Test the load-exclusive monitor is cleared on context switch)
Test REGRESSIONS0002 passed
Starting test 77: RETYPE0000
Running test RETYPE0000 (Retype test)
[0m[30;1m<<[0m[32mseL4(CPU 0)][0m[30;1m [decodeUntypedInvocation/147 T0xffffffff80041f0400 "27" @41e0a4]: Untyped

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Retype: Insufficient memory (1 * 2147483648 bytes needed, 8192 bytes available).>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeUntypedInvocation/208 T0xffffffff80041f0400 "27" @41e0a4]: Untyped
Retype: Insufficient memory (1 * 1073741824 bytes needed, 8192 bytes available).>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeUntypedInvocation/208 T0xffffffff80041f0400 "27" @41e0a4]: Untyped
Retype: Insufficient memory (1 * 536870912 bytes needed, 8192 bytes available).>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeUntypedInvocation/208 T0xffffffff80041f0400 "27" @41e0a4]: Untyped
Retype: Insufficient memory (1 * 268435456 bytes needed, 8192 bytes available).>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeUntypedInvocation/208 T0xffffffff80041f0400 "27" @41e0a4]: Untyped
Retype: Insufficient memory (1 * 134217728 bytes needed, 8192 bytes available).>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeUntypedInvocation/208 T0xffffffff80041f0400 "27" @41e0a4]: Untyped
Retype: Insufficient memory (1 * 67108864 bytes needed, 8192 bytes available).>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeUntypedInvocation/208 T0xffffffff80041f0400 "27" @41e0a4]: Untyped
Retype: Insufficient memory (1 * 33554432 bytes needed, 8192 bytes available).>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeUntypedInvocation/208 T0xffffffff80041f0400 "27" @41e0a4]: Untyped
Retype: Insufficient memory (1 * 16777216 bytes needed, 8192 bytes available).>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeUntypedInvocation/208 T0xffffffff80041f0400 "27" @41e0a4]: Untyped
Retype: Insufficient memory (1 * 8388608 bytes needed, 8192 bytes available).>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeUntypedInvocation/208 T0xffffffff80041f0400 "27" @41e0a4]: Untyped
Retype: Insufficient memory (1 * 4194304 bytes needed, 8192 bytes available).>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeUntypedInvocation/208 T0xffffffff80041f0400 "27" @41e0a4]: Untyped
Retype: Insufficient memory (1 * 2097152 bytes needed, 8192 bytes available).>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeUntypedInvocation/208 T0xffffffff80041f0400 "27" @41e0a4]: Untyped
Retype: Insufficient memory (1 * 1048576 bytes needed, 8192 bytes available).>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeUntypedInvocation/208 T0xffffffff80041f0400 "27" @41e0a4]: Untyped
Retype: Insufficient memory (1 * 524288 bytes needed, 8192 bytes available).>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeUntypedInvocation/208 T0xffffffff80041f0400 "27" @41e0a4]: Untyped
Retype: Insufficient memory (1 * 262144 bytes needed, 8192 bytes available).>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeUntypedInvocation/208 T0xffffffff80041f0400 "27" @41e0a4]: Untyped
Retype: Insufficient memory (1 * 131072 bytes needed, 8192 bytes available).>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeUntypedInvocation/208 T0xffffffff80041f0400 "27" @41e0a4]: Untyped
Retype: Insufficient memory (1 * 65536 bytes needed, 8192 bytes available).>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeUntypedInvocation/208 T0xffffffff80041f0400 "27" @41e0a4]: Untyped
Retype: Insufficient memory (1 * 32768 bytes needed, 8192 bytes available).>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeUntypedInvocation/208 T0xffffffff80041f0400 "27" @41e0a4]: Untyped
Retype: Insufficient memory (1 * 16384 bytes needed, 8192 bytes available).>>[0m
Test RETYPE0001 passed
Starting test 79: RETYPE0002
Running test RETYPE0002 (Incremental retype test #2)
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeUntypedInvocation/147 T0xffffffff80041f0400 "27" @41e0a4]: Untyped
Retype: Number of requested objects (1048576) too small or large.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeUntypedInvocation/169 T0xffffffff80041f0400 "27" @41e0a4]: Untyped
Retype: Slot #100 in destination window non-empty.>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeUntypedInvocation/208 T0xffffffff80041f0400 "27" @41e0a4]: Untyped
Retype: Insufficient memory (1 * 4096 bytes needed, 0 bytes available).>>[0m
Test RETYPE0002 passed
Starting test 80: SCHED0000
Running test SCHED0000 (Test suspending and resuming a thread (flaky))
Test SCHED0000 passed
Starting test 81: SCHED0002
Running test SCHED0002 (Test resuming ourselves)
Test SCHED0002 passed
Starting test 82: SCHED0003
Running test SCHED0003 (Test TCB suspend/resume)
Test SCHED0003 passed
Starting test 83: SCHED0004
Running test SCHED0004 (Test threads at all priorities)
Test SCHED0004 passed
Starting test 84: SCHED0005
Running test SCHED0005 (Test set priority)
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeSetPriority/1171 T0xffffffff8000682400 "child of: '27'" @43e0d8]:
TCB SetPriority: Requested priority 253 too high (max 251).>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeSetPriority/1171 T0xffffffff800066dc00 "child of: '27'" @43e0d8]:
TCB SetPriority: Requested priority 250 too high (max 249).>>[0m
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeSetMCPriority/1215 T0xffffffff800066dc00 "child of: '27'" @43e0d8]:
TCB SetMCPriority: Requested maximum controlled priority 253 too high (max 249).>>[0m
Test SCHED0005 passed
Starting test 85: SCHED0006
Running test SCHED0006 (Test IPC priorities for Send)
Test SCHED0006 passed
Starting test 86: SCHED0020
Running test SCHED0020 (test set prio to a higher prio runs higher prio thread)

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Test SCHED0020 passed
Starting test 87: SCHED0021
Running test SCHED0021 (Test for pre-emption during running of many threads with equal prio)
Test SCHED0021 passed
Starting test 88: SERSERV_CLIENT_001
Running test SERSERV_CLIENT_001 (Connect from client threads)
Test SERSERV_CLIENT_001 passed
Starting test 89: SERSERV_CLIENT_002
Running test SERSERV_CLIENT_002 (Printf from client threads)
[0m[33mHello, world!
[0mTest SERSERV_CLIENT_002 passed
Starting test 90: SERSERV_CLIENT_003
Running test SERSERV_CLIENT_003 (Write from client threads)
[0m[33mHello, world!
[0mTest SERSERV_CLIENT_003 passed
Starting test 91: SERSERV_CLIENT_004
Running test SERSERV_CLIENT_004 (Printf, then write, then reset connection, and Printf, then write again, from
client threads)
[0m[33mHello, world!
[0m[0m[33mHello, world!
[0m[0m[34mHello, world!
[0m[0m[34mHello, world!
[0mTest SERSERV_CLIENT_004 passed
Starting test 92: SERSERV_CLIENT_005
Running test SERSERV_CLIENT_005 (Connect, then disconnect from server on all threads, then kill server from
parent thread)
Test SERSERV_CLIENT_005 passed
Starting test 93: SERSERV_CLI_PROC_001
Running test SERSERV_CLI_PROC_001 (Connect to server from a client in another VSpace and CSpace)
Test SERSERV_CLI_PROC_001 passed
Starting test 94: SERSERV_CLI_PROC_002
Running test SERSERV_CLI_PROC_002 (Connect to server and printf(, true), from a client in another VSpace and
CSpace)
[0m[33mHello, world!
[0mTest SERSERV_CLI_PROC_002 passed
Starting test 95: SERSERV_CLI_PROC_003
Running test SERSERV_CLI_PROC_003 (Connect to server and write(, true), from a client in another VSpace and
CSpace)
[0m[33mHello, world!
[0mTest SERSERV_CLI_PROC_003 passed
Starting test 96: SERSERV_CLI_PROC_004
Running test SERSERV_CLI_PROC_004 (Connect to server, printf(), write(, true), then disconnect, then reconnect
and printf() and write() again, from clients in other VSpaces and CSpaces)
[0m[33mHello, world!
[0m[0m[33mHello, world!
[0m[0m[34mHello, world!
[0m[0m[34mHello, world!
[0mTest SERSERV_CLI_PROC_004 passed
Starting test 97: SERSERV_CLI_PROC_005
Running test SERSERV_CLI_PROC_005 (Connect to server then disconnect on all clients (in different VSpace/CSpace
containers), and finally kill the server from the parent)
Test SERSERV_CLI_PROC_005 passed
Starting test 98: SERSERV_PARENT_001
Running test SERSERV_PARENT_001 (Serial server spawn test)
Test SERSERV_PARENT_001 passed
Starting test 99: SERSERV_PARENT_002
Running test SERSERV_PARENT_002 (Test connecting to the server from a parent thread)
Test SERSERV_PARENT_002 passed
Starting test 100: SERSERV_PARENT_003
Running test SERSERV_PARENT_003 (Printf() from a connected parent thread)
[0m[33mHello, world!
[0mTest SERSERV_PARENT_003 passed
Starting test 101: SERSERV_PARENT_004
Running test SERSERV_PARENT_004 (Write() from a connected parent thread)
[0m[33mHello, world!
[0mTest SERSERV_PARENT_004 passed
Starting test 102: SERSERV_PARENT_005
Running test SERSERV_PARENT_005 (Test printf() and write() from a parent thread when after a connection reset
(disconnect/reconnect))
[0m[33mHello, world!
[0m[0m[33mHello, world!

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[0m[0m[33mHello, world!
[0m[0m[33mHello, world!
[0mTest SERSERV_PARENT_005 passed
Starting test 103: SERSERV_PARENT_006
Running test SERSERV_PARENT_006 (Kill the Server from the Parent thread)
Test SERSERV_PARENT_006 passed
Starting test 104: SERSERV_PARENT_007
Running test SERSERV_PARENT_007 (Test a series of unexpected input values to spawn_thread)
Test SERSERV_PARENT_007 passed
Starting test 105: SERSERV_PARENT_008
Running test SERSERV_PARENT_008 (Test a series of unexpected input values to connect())
Test SERSERV_PARENT_008 passed
Starting test 106: SERSERV_PARENT_009
Running test SERSERV_PARENT_009 (Test a series of unexpected input values to printf())
serial_server_printf@clientapi.c:178 Serserv Client: printf: NULL passed for required arguments.
 Is connection handle valid?
serial_server_printf@clientapi.c:178 Serserv Client: printf: NULL passed for required arguments.
 Is connection handle valid?
Test SERSERV_PARENT_009 passed
Starting test 107: SERSERV_PARENT_010
Running test SERSERV_PARENT_010 (Test a series of unexpected input values to write())
serial_server_write@clientapi.c:221 Serserv Client: printf: NULL passed for required arguments.
 Is connection handle valid?
serial_server_write@clientapi.c:221 Serserv Client: printf: NULL passed for required arguments.
 Is connection handle valid?
Test SERSERV_PARENT_010 passed
Starting test 108: SYNC001
Running test SYNC001 (libsel4sync Test binary semaphores)
Test SYNC001 passed
Starting test 109: SYNC002
Running test SYNC002 (libsel4sync Test semaphores)
Test SYNC002 passed
Starting test 110: SYNC003
Running test SYNC003 (libsel4sync Test monitors)
Test SYNC003 passed
Starting test 111: SYNC004
Running test SYNC004 (libsel4sync Test monitors - broadcast)
Test SYNC004 passed
Starting test 112: THREADS0004
Running test THREADS0004 (sel4_TCB_Configure with a NULL CSpace should fail)
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeTCBConfigure/1106 T0xffffffff80041f0400 "27" @44ffb4]: TCB
Configure: CSpace cap is invalid.>>[0m
Test THREADS0004 passed
Starting test 113: THREADS0005
Running test THREADS0005 (sel4_TCB_SetSpace with a NULL CSpace should fail)
[0m[30;1m<<[0m[32mseL4(CPU 0)[0m[30;1m [decodeSetSpace/1498 T0xffffffff80041f0400 "27" @44ffb4]: TCB SetSpace:
Invalid CNode cap.>>[0m
Test THREADS0005 passed
Starting test 114: TLS0001
Running test TLS0001 (Test root thread accessing __thread variables)
Test TLS0001 passed
Starting test 115: TLS0002
Running test TLS0002 (Test multiple threads using __thread variables)
Test TLS0002 passed
Starting test 116: TLS0006
Running test TLS0006 (sel4utils_thread with distinct TLS should not interfere)
Test TLS0006 passed
Starting test 117: TRIVIAL0000
Running test TRIVIAL0000 (Ensure the test framework functions)
Test TRIVIAL0000 passed
Starting test 118: TRIVIAL0001
Running test TRIVIAL0001 (Ensure the allocator works)
Test TRIVIAL0001 passed
Starting test 119: TRIVIAL0002
Running test TRIVIAL0002 (Ensure the allocator works more than once)
Test TRIVIAL0002 passed
Starting test 120: VSPACE0000
Running test VSPACE0000 (Test threads in different cspace/vspace)
Test VSPACE0000 passed
Starting test 121: VSPACE0001
Running test VSPACE0001 (Test unmapping a page after deleting the PD)

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Test VSPACE0001 passed
Starting test 122: VSPACE0002
Running test VSPACE0002 (Test create ASID pool)
Test VSPACE0002 passed
Starting test 123: VSPACE0003
Running test VSPACE0003 (Test create multiple ASID pools)
Test VSPACE0003 passed
Starting test 124: VSPACE0004
Running test VSPACE0004 (Test running out of ASID pools)
Test VSPACE0004 passed
Starting test 125: VSPACE0005
Running test VSPACE0005 (Test overassigning ASID pool)
Test VSPACE0005 passed
Starting test 126: VSPACE0006
Running test VSPACE0006 (Test touching all available ASID pools)
Test VSPACE0006 passed
Starting test 128: Test all tests ran
Test suite passed. 128 tests passed. 41 tests disabled.
All is well in the universe
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